

1 The question

Long-term bond yields are anchored by beliefs about where short rates settle in the long run. A revision to that endpoint passes into the yield at every maturity, undamped by horizon, in a way that news about the current policy stance does not. This is why the endpoint has been at the center of the term-structure literature since Kozicki and Tinsley’s shifting endpoints, and why Bauer and Rudebusch’s falling stars put a slow-moving anchor at the heart of it. What is unsettled is where those beliefs come from, and whether the central bank is one of the sources. Consider a policy rule written as follows,

$$i_t = \underbrace{r_t^* + \pi_t^*}_{\text{intercept}} + \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t, \quad (1)$$

r^* is the *intercept* of the reaction function, so “the Fed reveals r^* ” means the Fed is telling markets about the constant in its own rule. Bauer, Pflueger & Sunderam (2024) estimate the perceived *slope* ϕ_π and show it prices term premia; they absorb the intercept into forecaster fixed effects. The intercept has been assumed known, assumed common, or differenced away.

How does the Fed move beliefs about the intercept of its own reaction function, and what does disagreement about that intercept do to the price of long bonds?

In three parts:

1. **Who moves whom:** Does the FOMC’s published longer-run projection move primary dealers’ beliefs about the same object, and by how much? And does the dealers’ projection, which the Desk reports to the Committee before every meeting, move the FOMC’s own, and by how much?
2. **Pricing:** Does a revision to the published projection move the yield curve with the maturity profile of an endpoint shock? Can the term-premium response be separated from stance news and from a perceived policy error?
3. **Disagreement:** Does dispersion in beliefs about the intercept carry a risk premium whose sensitivity does not decay with maturity? Does a signed Fed-minus-market gap move the level of yields, with the split between breakeven and real-rate responses pinned by the perceived rule?

2 Decomposing Long Yields

Start from the definition of a long yield as an average of expected short rates plus a residual:

$$y_t^{(n)} = \underbrace{\frac{1}{n} \sum_{j=0}^{n-1} E_t[i_{t+j}]}_{\text{expectations component } rn_t^{(n)}} + TP_t^{(n)}. \quad (2)$$

Now split the nominal policy rate into a neutral level and a deviation from it. Define the *neutral nominal rate* and the *policy gap*

$$i_t^* \equiv r_t^* + \pi_t^*, \quad \text{gap}_t \equiv i_t - i_t^* = \underbrace{(i_t - E_t \pi_{t+1} - r_t^*)}_{\text{real policy stance}} + \underbrace{(E_t \pi_{t+1} - \pi_t^*)}_{\text{inflation deviation}}, \quad (3)$$

where r_t^* is the equilibrium real rate and π_t^* the long-run inflation anchor. The gap is therefore **the stance of policy**: how far the Fed is holding the short rate above or below the level that would be neutral, plus how far inflation is expected to run from its anchor. Under a Taylor-type rule $i_t = i_t^* + \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t$, the gap is exactly the systematic and discretionary response to the cycle,

$$\text{gap}_t = \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t. \quad (4)$$

Substituting (3) into (2), and adding the martingale assumption for the endpoints — $E_t[r_{t+j}^*] = r_t^*$ and $E_t[\pi_{t+j}^*] = \pi_t^*$ — gives

$$y_t^{(n)} = r_t^* + \pi_t^* + \frac{1}{n} \sum_{j=0}^{n-1} E_t[\text{gap}_{t+j}] + TP_t^{(n)}. \quad (5)$$

Equivalently, in terms of the neutral nominal rate,

$$y_t^{(n)} = i_t^* + \frac{1}{n} \sum_{j=0}^{n-1} E_t[\text{gap}_{t+j}] + TP_t^{(n)}. \quad (6)$$

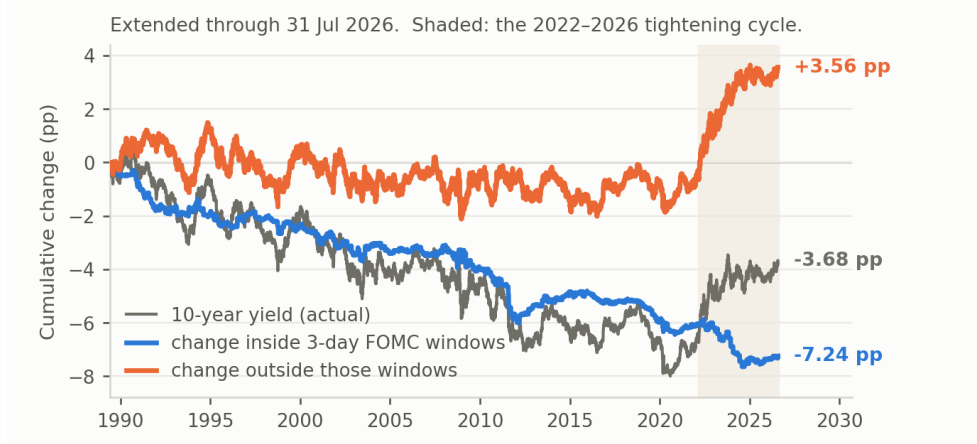


Figure 1: Fact 1. Cumulative change in the ten-year yield from 2 June 1989, split into the part accruing inside three-day FOMC windows and the part accruing on all other days. Shaded: the 2022–2026 tightening cycle.

3 Three facts

3.1 Fact 1: through 2021, yield changes outside FOMC windows net to zero

Replicating Hillenbrand (2025) on GSW yields and a hand-built event list: yields move on non-FOMC days about as much as on FOMC days — daily standard deviations of 5.60 against 6.72 bp. What differs is that outside the window the moves offset. Across 7,035 non-FOMC days the cumulative change is -0.82 pp, a drift of -0.012 bp/day with $t = -0.18$; inside three-day FOMC windows, 11.8% of trading days, the ten-year fell 6.16 pp of a total 6.99. Hillenbrand’s explanation is **long-run Fed guidance**: announcements are the moments at which the Fed reveals its own estimate of where rates settle in the long run — implicitly before 2012, when the rate decision itself disclosed the Committee’s read of the natural rate, and explicitly after 2012 through the SEP longer-run dot.

$$y_t^{(n)} = i_t^* + \frac{1}{n} \sum_{j=0}^{n-1} E_t[\text{gap}_{t+j}] + TP_t^{(n)},$$

Hillenbrand’s reading is that window moves are Δi^* , which requires the other two terms to contribute little. For the stance he appeals to the far-forward evidence: the same downward trend appears in the 5-year, 5-year-forward rate, where the current stance should long since have washed out. For the term premium he reports in the Internet Appendix that model-based estimates *do* fall in these windows, and sets the finding aside on the grounds that those models cannot represent a secular decline. A third result, that the decline since 1999 is in TIPS rather than breakevens, does not exclude a term but tells us the composition of Δi^* : it is the real endpoint moving, not the inflation anchor.

3.2 Fact 2: the two beliefs move together, and asymmetrically

The FOMC publishes its longer-run funds rate projection quarterly; the NY Fed surveys primary dealers for the same object about two weeks before every meeting. Netting the 2% objective from both gives two comparable measures of r^* (Figure 2). Across 48 matched meetings (46–47 for the revision regressions): the gap has a standard deviation of 0.105 pp and a range of -0.11 to $+0.30$, exceeding 25 bp at 4% of meetings; dealers move with the Fed nearly one-for-one ($\beta = 0.82$, $t = 6.86$); the reverse relation is present but weaker ($\beta = 0.31$, $t = 3.91$). The asymmetry is not

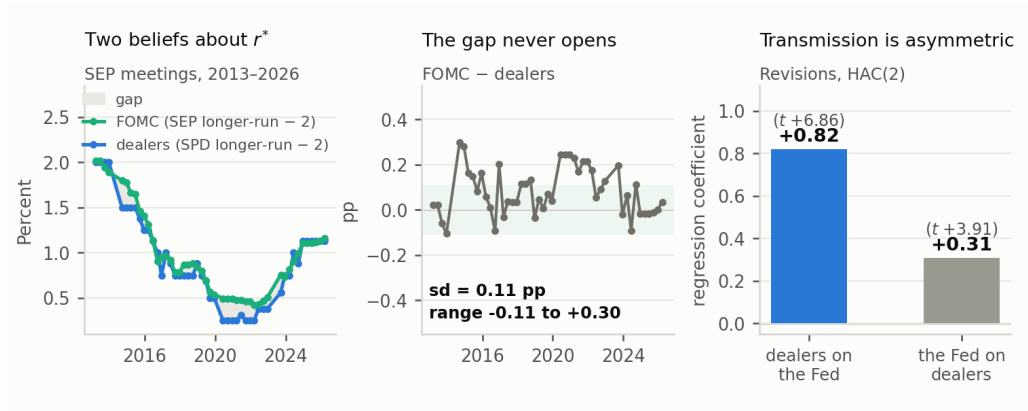


Figure 2: Fact 2. Left: the FOMC’s longer-run projection and the primary dealers’ longer-run funds rate, each less 2.0. Centre: the difference. Right: revision regressions, HAC(2), 46–47 meetings.

mechanical — the Desk briefs the FOMC with survey results before each meeting — but it is not a clean lead-lag either. Dealer revision loads on both the preceding dot revision (0.60) and the contemporaneous one (0.58), so part of what looks like following is anticipation; and intermeeting macro news moves both series, so these regressions describe timing, not identified transmission. Both series are observed eight times a year. Section 5 adds a pricing counterpart to this fact: the announcement window prices exactly the part of the dot revision that the dealers’ own preceding revision did not anticipate.

3.3 Fact 3: the FOMC’s r^* and the econometric r^* move in opposite directions

The FOMC’s estimate can be compared against the leading model-based alternative. Holston, Laubach & Williams estimate the equilibrium real rate from a semi-structural macro model with a Kalman filter, quarterly from 1961. Under the martingale assumption, these two are estimates of the same object: HLW’s contemporaneous r_t^* and the FOMC’s longer-run projection coincide when $E_t[r_{t+j}^*] = r_t^*$. They do not behave like estimates of the same object (Figure 3). Across the 57 SEP meetings the correlation of levels is -0.93 . Between January 2012 and March 2022 the FOMC marked its own r^* down by 1.78 pp, from 2.21% to 0.42%, while HLW marked r^* up by 1.37 pp, from 0.89% to a peak of 2.26%. After 2022 they swap: the FOMC revises up 0.74 pp while HLW falls back. The two series cross exactly once, in June 2016, and the mean absolute gap between them is 0.88 pp — they sit within 25 bp of each other at only 5% of meetings. Three things follow. First, the Fed’s r^* is not easily recoverable from an econometric model. Second, the same holds for the market side: it is hard to justify a model-based endpoint to serve as the market perception of r^* . Third, the premise that the Fed holds an information advantage about r^* has an observable consequence: over the decade in which the Committee cut its estimate by 1.78 pp, the filtered econometric estimate moved the other way. The comparison uses one-sided estimates under current parameters; whether it survives on the real-time vintages, which declined over 2012–2019, and on the post-COVID revision of the model is being verified. One candidate explanation can be rejected immediately. HLW estimates r^* from a model; the Fed’s is inferred from its reported i^* and π^* , and the SEP reports $\pi^* = 2.0\%$ with zero variation over the observed sample. If participants report i^* sincerely but hold π^* at 2% to anchor expectations, the netting $r_{\text{Fed}}^* = \text{dot} - 2$ would mislabel the Fed’s real endpoint and the two series would not be comparable. Figure 4 tests this directly: it computes the long-run inflation expectation that would have to be embedded in the FOMC’s projection for the two estimates to agree. That implied series runs from 3.46% in September 2012

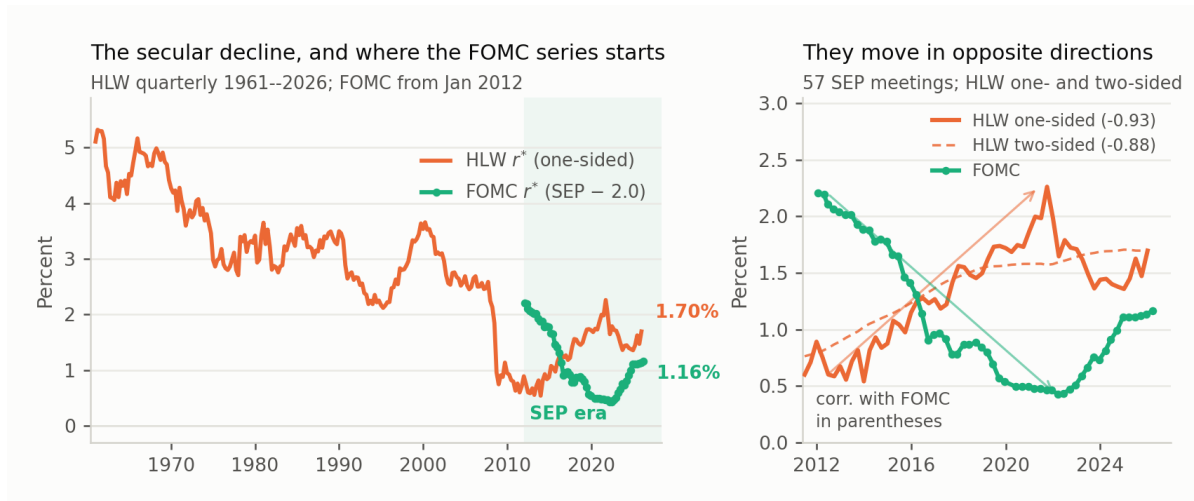


Figure 3: Fact 3. Left: HLW r^* over the full quarterly sample with the FOMC series, which begins in January 2012, overlaid; the SEP era is shaded. Right: the overlap sample, with arrows marking the direction each series travels over 2012–2022.

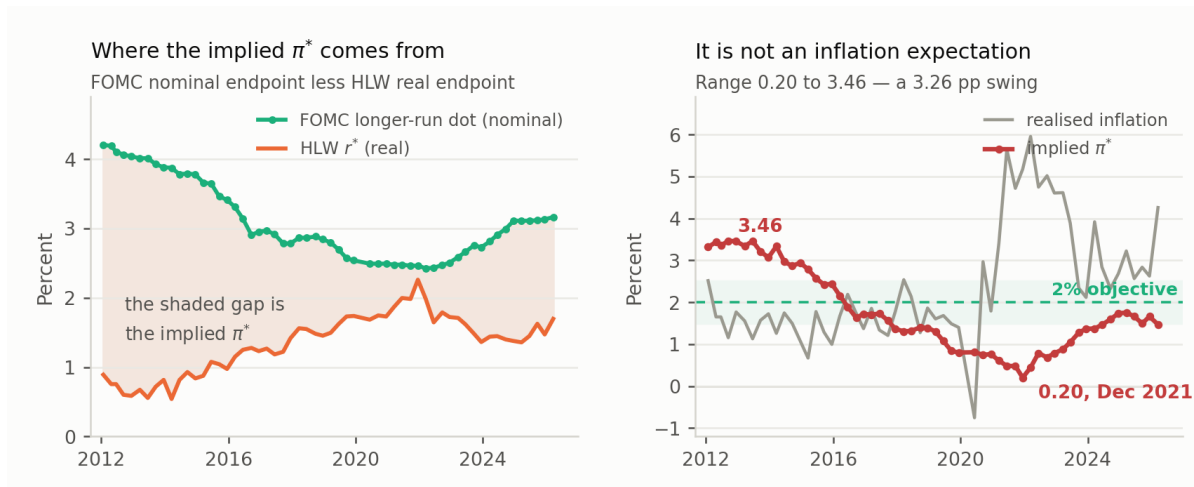


Figure 4: The inflation expectation needed to reconcile the two estimates. Left: the FOMC’s longer-run federal funds projection, which is nominal, against HLW’s one-sided r^* , which is real, at the 57 SEP meetings; the shaded area between them is the long-run inflation expectation that would have to be embedded in the FOMC’s projection for the two series to be mutually consistent. Right: that implied series plotted against realised inflation, with the 2% objective marked and a 1.5–2.5% band shaded. The implied series lies outside the 1.5–2.5% band at 75% of meetings. Realised inflation is HLW’s own four-quarter input series rather than headline PCE.

to 0.20% in December 2021 — at its trough it requires FOMC participants to have expected 0.20% long-run inflation at a moment when realised inflation was 5.17%. Whatever separates the two estimates, it is not the 2% netting.

4 Framework: intercept disagreement in an opinionated-markets model

Caballero & Simsek (2022) study a Fed and a market that disagree, each confident in its own view, with the Fed setting policy on its belief and the market pricing assets knowing it will. The market therefore anticipates policy “mistakes” and prices the resulting risk. Their state variable is aggregate demand. This section runs the same logic with the disagreement located in the *intercept of the policy rule*. What distinguishes intercept disagreement is not its persistence — the estimated Fed–dealer gap mean-reverts with a half-life of about six months — but that it maps into a *composition* restriction across the nominal and real curves that demand disagreement does not generate.

4.1 Setup

Both parties know the rule and observe the state. They differ only in the neutral real rate: the Fed believes r^{*F} , the market believes r^{*m} , and

$$\delta_t \equiv r_t^{*m} - r_t^{*F}. \quad (7)$$

The Fed sets policy on its own estimate,

$$i_t = r_t^{*F} + \pi^* + \phi_\pi(\pi_t - \pi^*) + \phi_x x_t, \quad \phi_\pi > 1, \quad (8)$$

and the market evaluates the resulting stance against *its* neutral rate, perceiving policy to be δ_t looser than the Fed intends.

4.2 Three roles, and whose belief enters where

	Role	Estimate used
Fed	Sets i_t through the rule (8)	r^{*F}
Economy	Adjusts π until the real rate reaches the natural rate	the truth, r^*
Market	Forecasts the path of i ; prices bonds off that forecast	r^{*m}

The market’s error is that it substitutes its own r^{*m} for the second row when it runs the economy forward in its head. Long yields follow by averaging the expected short-rate path the market computes this way; the resulting pricing equation is (13) below.

4.3 The market’s subjective steady state

Consider the steady state the market *forecasts*, holding its belief dogmatically. Two conditions define it. Over the long-run the Fed closes the output gap, so $\phi_x x_t$ drops out and permanent accommodation is absorbed entirely by inflation rather than by output; and the natural rate is by definition the real rate consistent with a zero gap, so $i - \pi$ equals the natural rate — which, in the market’s forecast, is the one the market believes in: $i - \pi = r^{*m}$. Substituting (8),

$$r^{*m} = r^{*F} + \pi^* + \phi_\pi(\pi - \pi^*) - \pi \implies (\phi_\pi - 1)(\pi - \pi^*) = \delta,$$

so the market expects inflation to settle permanently away from target at

$$\boxed{\pi - \pi^* = \frac{\delta_t}{\phi_\pi - 1}} \quad (9)$$

The target π^* is a parameter of the rule, set by the Fed by announcement and held at 2% throughout the sample — all 594 archived longer-run PCE submissions are exactly 2.0, with zero within-meeting

dispersion. What moves is *realised* long-run inflation π . The Fed does not revise its target; it persistently misses one it continues to hold, because it is setting policy against the wrong intercept. The adjustment runs through the real rate: at $\pi = \pi^*$ the real rate is r^{*F} , which is below the natural rate, so demand exceeds supply and inflation rises; each point of inflation above target raises i by ϕ_π and therefore the real rate by $\phi_\pi - 1$; this continues until the real rate has climbed to meet the natural rate. Nobody chooses the natural rate, and nobody chooses π — inflation is the free variable.

Subjective, not actual. In the *true* steady state $i - \pi = r^*$, and (9) with δ replaced by $r^* - r^{*F}$ gives the realised bias. The market's version coincides with it only if the market happens to be right. Nothing below requires that it is: bonds are priced off $E_t^m[i_{t+j}]$, which the market computes using its own intercept at every step, so (9) and (10) are statements about market expectations and hold whether or not those expectations are correct.

The role of the Taylor principle. Under true dynamics the real rate is $r^{*F} + (\phi_\pi - 1)(\pi - \pi^*)$, so inflation rises until the second term offsets the Fed's intercept error, and it converges to a finite bias only because $\phi_\pi > 1$. The macro logic is Orphanides' account of the Great Inflation — a policy rule fed a mismeasured real-time natural rate producing a sustained inflation bias. The analogy is structural rather than literal: Orphanides locates the mismeasurement in the output gap and the natural rate of unemployment, i.e. in the $\phi_x x_t$ term, whereas the mechanism here is in the intercept. Orphanides & Williams on natural-rate misperception under learning is the closer antecedent.

4.4 The amplification result, and what it does and does not identify

The nominal endpoint the market prices is $r^{*m} + \pi^* + \delta/(\phi_\pi - 1)$; the endpoint the FOMC publishes is $r^{*F} + \pi^*$. The difference is

$$\underbrace{i_t^{*m} - i_t^{*F}}_{\text{endpoint gap}} = \frac{\phi_\pi}{\phi_\pi - 1} \delta_t \quad (10)$$

Identification. Given the Fed's published i^{*F} , the market's expected terminal nominal rate reveals r^{*m} — but only up to the perceived ϕ_π . Equation (10) is one equation in two unknowns: the same observed endpoint gap is consistent with a small δ and a weakly-responding perceived Fed, or a large δ and an aggressive one. Recovering the perceived ϕ_π from the ratio therefore requires observing δ and the endpoint gap *separately*, and on current data that is not possible. The SPD's longer-run PCE median is 2.0 in every one of the 91 surveys in the sample, so dealers condition on inflation at target and the survey gap measures δ itself; the amplified endpoint gap does not appear in the survey. Nor can it be taken from market prices: far-forward risk-neutral rates load heavily on the current short rate and cannot serve as an independent endpoint. The result is therefore best read as an identification statement: the published dot and the market's terminal rate differ by a *multiple* of the underlying disagreement, and the multiple is the perceived Taylor coefficient. This statement does not depend on the calibration below.

The magnitude claim. Read as a statement about levels, a small disagreement about r^* implies a large disagreement about where nominal rates would settle, with the multiplier set by the perceived aggressiveness of the rule:

ϕ_π	inflation bias per 1 pp of δ	endpoint gap per 1 pp of δ	endpoint gap when $\text{sd}(\delta) = 0.105$
1.25	4.00	5.00	52 bp
1.50	2.00	3.00	32 bp
2.00	1.00	2.00	21 bp
2.50	0.67	1.67	18 bp

These are steady-state numbers: the multiplier is attained only if the market expects the disagreement never to be resolved. At the estimated resolution speed — the Fed–dealer gap has an AR(1) coefficient of 0.849 per meeting, a half-life of about six months — the ten-year loading falls from its asymptotic value of 3.0 to roughly 0.1 unless about half of any given disagreement is permanent (§4.5). The table is an upper bound, not a statement about observed magnitudes.

The steady state is self-defeating. Reaching it requires the Fed to observe inflation running $\delta/(\phi_\pi - 1)$ above target — at $\phi_\pi = 1.5$, twice the size of δ itself — indefinitely, and never to revise r^{*F} . But a persistent inflation gap is precisely the signal that identifies an intercept error. The mechanism therefore generates the evidence that resolves the disagreement it depends on. This objection is prior to and independent of the estimated persistence, and it is a further reason to expect the permanent component of disagreement to be small.

4.5 The path to the steady state

The steady-state gap in (10) is the limit of a path. Along the market’s forecast, the inflation bias builds as the economy responds to the too-low real rate, and the disagreement itself shrinks as the Fed revises r^{*F} toward the market’s view; a share α of any given disagreement is never resolved. Parameterising the build at rate κ and the resolution at rate μ , the market’s expected short rate at horizon j loads on the current disagreement as

$$\lambda_j = \frac{\phi_\pi}{\phi_\pi - 1} \underbrace{(1 - e^{-\kappa j})}_{\text{bias builds}} \left[\underbrace{\alpha}_{\text{unresolved}} + \underbrace{(1 - \alpha)(1 - \mu)^j}_{\text{Fed learns}} \right], \quad \alpha \in [0, 1]. \quad (11)$$

The two polar cases anchor the interpretation. At $\alpha = 1$ the loading rises monotonically with horizon and converges to the multiplier in (10): the amplification result is the long-horizon limit of (11). At $\alpha = 0$ the loading is hump-shaped and converges to zero: under pure learning the amplified endpoint is a transient that has largely disappeared by the maturities at which it would be priced. The two claims of this section — a large steady-state gap and a falsifiable hump in maturity — therefore trade off against each other through α .

Calibration. μ is estimated from the gap’s own persistence: an AR(1) coefficient of 0.849 per meeting (half-life 4.2 meetings, or 6.4 months) gives $\rho_{\text{month}} = 0.897$ and $\hat{\mu} = 0.103$. Setting $\kappa = 0.10$ per month and $\phi_\pi = 1.5$, the averaged loading $\Lambda(n) = n^{-1} \sum_{j < n} \lambda_j$ — which is the coefficient a maturity regression estimates, and equals the ten-year endpoint multiplier at $n = 10\text{y}$ — is:

α	$\Lambda, 2\text{y}$	$\Lambda, 5\text{y}$	$\Lambda, 10\text{y}$
0.00	0.47	0.22	0.11
0.10	0.60	0.44	0.37
0.25	0.80	0.78	0.77
0.50	1.14	1.35	1.42
1.00	1.81	2.48	2.74
asymptote ($n \rightarrow \infty$)			3.00

At $\alpha = 0$ — pure learning at the estimated speed — the ten-year loading is 0.11: a 10 bp δ moves the ten-year expectations component by about 1 bp, against 30 bp in the steady state. Delivering even half the asymptotic multiplier requires $\alpha \approx 0.5$. Whether the amplification is quantitatively relevant therefore turns entirely on the permanent share of disagreement, and if α cannot be bounded away from zero the magnitude claim should be dropped rather than reported at its steady-state value. §5.6 now bounds it from both sides: the far-forward evidence puts α at about 0.25, with a 95% interval of roughly $[0.10, 0.45]$, rejecting the steady-state multiplier and pure transience alike.

Peak location. With $m \equiv -\ln(1 - \mu)$ and $\alpha = 0$, λ_j peaks at

$$j^* = \frac{1}{\kappa} \ln\left(1 + \frac{\kappa}{m}\right), \quad (12)$$

which is 6.5 months at the calibration above. The object a maturity regression estimates is Λ , not λ , and a running average peaks strictly later, where $\lambda_n = \Lambda(n)$: at 12 months for $\alpha = 0$ and 28 months for $\alpha = 0.25$, moving out and flattening as α rises. Both κ and μ are estimable outside the event sample, so the location of the peak is a quantitative prediction rather than an asserted shape — though the test loses power as α grows, since the hump flattens toward the monotone case.

4.6 Pricing

Averaging the market’s expected short rates gives

$$y_t^{(n)} = \underbrace{i_t^{*F}}_{\text{published endpoint}} + \underbrace{\omega(\rho, n) \text{ gap}_t}_{\text{stance}} + \underbrace{\Lambda(n) \delta_t}_{\text{perceived error}} + \underbrace{TP^{(n)}(\sigma_{\delta, t}^2)}_{\text{disagreement premium}}. \quad (13)$$

The yield is anchored on the Fed’s *published* endpoint. This is not an assumption about whose belief is correct; it follows from the roles above. Realised short rates are generated by the Fed’s rule, so the path the market forecasts starts at the Fed’s intercept, and the market’s own belief enters only through the inflation path it expects that rule to produce — the $\Lambda(n) \delta_t$ term. The two anchors are reconciled at the long end: when $\alpha = 1$, $\Lambda(n) \rightarrow \phi_\pi / (\phi_\pi - 1)$ and the long yield converges to the market’s terminal rate; when $\alpha < 1$, the Fed’s expected learning pulls it back toward the published endpoint. Every regressor in (13) is observable: i^{*F} is published, the stance is measurable, and δ is the survey gap.

4.7 Predictions

1. **Amplification: not directly measurable on current data.** As established above, (10) is one equation in two unknowns, and the survey provides δ but not the amplified endpoint gap. If an independent measure of the market’s terminal rate becomes available, the ratio identifies the market’s perceived ϕ_π , directly comparable to the survey-based estimates in Bauer, Pflueger & Sunderam (2024). Until then, the amplification enters the empirical work only through its implications for the loadings in (13).
2. **A composition restriction on the TIPS split.** The endpoint gap divides into δ in the real dimension and $\delta/(\phi_\pi - 1)$ in the inflation dimension, so

$$\frac{\text{real endpoint gap}}{\text{breakeven gap}} = \phi_\pi - 1.$$

At $\phi_\pi = 1.5$ breakevens move twice as much as real forwards, *in the same direction*. Demand disagreement generates no such restriction, which is what distinguishes the intercept location.

The restriction is testable on responses rather than levels: regressing far-forward real and breakeven window changes on $\Delta\delta$ and testing that the coefficient ratio equals $\phi_\pi - 1$ requires no decomposition of either curve. The binding constraint is power: with $\text{sd}(\delta) = 0.105$ and a six-month half-life, $\Delta\delta$ has little variance, and a power calculation must precede the test. *That calculation is in §5.2 and the test is run in §5.5. Two amendments follow from it. The ratio cancels Λ , so it is free of α — which makes this, not Prediction 5, the framework’s cleanest α -free content. And it must be estimated on dM between meetings rather than on $\Delta\delta$ in the announcement window, where the published-endpoint channel dominates.*

3. **Sign, and the absence of asymmetry.** $\delta > 0$ — the Fed’s r^* below the market’s — means policy is perceived too easy, so both the inflation bias and the endpoint gap are positive; $\delta < 0$ reverses both. This is linearity in δ , not sign-dependence: every expression in (9)–(13) is proportional to δ , so upward and downward disagreements are mirror images. Caballero & Simsek’s premium is genuinely sign-dependent; this model is not, and the differentiation from them rests on the maturity profile and the composition ratio in Prediction 2 rather than on any asymmetry. Generating asymmetry would require a nonlinearity — an occasionally-binding lower bound, or state-dependent ϕ_π — which is not in the present specification. *Tested in §5.7: symmetry is not rejected ($p = 0.79$ – 0.98).*
4. **Maturity profile.** The three terms of (13) carry three distinguishable shapes: a revision to the published endpoint i^{*F} loads flat, at one, across maturity; the stance term $\omega(\rho, n)$ declines monotonically; and the disagreement term $\Lambda(n)$ is hump-shaped for α small and rises monotonically toward $\phi_\pi/(\phi_\pi - 1)$ for α large, with the peak pinned by (κ, μ, α) as in §4.5. *Tested in §5.4: flat-at-one is rejected at the far end — the ten-year-forward pass-through of a dot revision is about one half, and §5.5 shows that half is entirely real.*
5. **State dependence, and a link to the perceived-slope literature.** The multiplier $\phi_\pi/(\phi_\pi - 1)$ is decreasing in ϕ_π : *the same r^* disagreement matters more when markets believe the Fed responds weakly.* Intercept disagreement and slope perception are complements, not rival explanations — a testable interaction, and the one prediction here that does not depend on α , since it concerns the shape of the multiplier in ϕ_π rather than its level. *Not yet testable: it needs a perceived- ϕ_π series. Note that Prediction 2’s ratio is also α -free, since Λ cancels from it.*
6. **The premium.** Uncertainty about δ propagates into the whole expected path undamped by horizon, so term-premium sensitivity to σ_δ^2 should rise with maturity and not decay — unlike stance uncertainty, which is damped by ω . This requires σ_δ^2 itself to be persistent even where δ mean-reverts; the two are separate assumptions. The nearest existing results are Cao, Crump, Eusepi & Moench, who show that forecaster dispersion about the long-run level of rates accounts for roughly a third of term-premium variation, and Cieslak, McMahon & Pang, who show that perceived policy mistakes move term premia rather than expected rates. The differentiation here is that the disagreement is Fed-versus-market rather than forecaster-versus-forecaster, and is located in the intercept of the rule, which is what generates the composition restriction in Prediction 2. *A proxy test in §5.8 finds nothing; the proxy is weak, and the prediction remains open.*

4.8 What the model assumes, stated plainly

The permanent share α . The magnitude of the amplification lives or dies on it, and the maturity-profile test loses power as it rises. In principle α is the unit-root share of the Fed–dealer gap; in practice, roughly 48 observations of a series with a standard deviation of 0.105 are unlikely to bound

it tightly, so the calibration table in §4.5 should be read as a sensitivity analysis rather than a set of point predictions. *The Taylor principle.* Everything here requires $\phi_\pi > 1$, and the multiplier diverges as $\phi_\pi \rightarrow 1$. That is not a defect — the model is saying that intercept errors are catastrophic for a weakly-responding central bank — but it means the empirical work must bound ϕ_π away from one, and estimates near the boundary count as evidence against the specification rather than as very large effects. *Measurement of δ .* The survey side conditions on inflation at target — the SPD’s longer-run PCE median is 2.0 in all 91 surveys — so the SPD gap measures δ directly. That validates it as the regressor in Prediction 2, and it is also why the amplified endpoint gap cannot be measured from the same source. *Linearity.* The model generates no asymmetry. The case for locating the disagreement in the intercept rests on the composition restriction and the maturity profile, not on sign-dependence.

5 The predictions, taken to the data

This section reports the first pass of the empirical program of §4.7 on the archive behind Facts 1–3: the SEP dot-plot panel, the SPD/SMP survey archive, the ACM daily decomposition, and the Gürkaynak–Sack–Wright nominal and TIPS curves, over the 48 matched meetings of Fact 2 (2013–2026). Five of the six predictions can now be confronted with data; Prediction 5 waits on a perceived- ϕ_π series. One result was not predicted at all, and it is the most useful: the announcement window prices the *unanticipated component of the dot*, measured against the dealers’ own preceding revision, and prices essentially nothing else.

5.1 Design: two revisions with different timing

Write $\delta_t = r_t^{*m} - r_t^{*F}$ from the Fact-2 panel: the Fed side is the mean of the longer-run dots released *at* meeting t , the market side the SPD/SMP longer-run median from the survey fielded roughly two weeks before. (This build reproduces Fact 2 exactly: 48 matched meetings, gap standard deviation 0.105, revision regressions 0.82 (t 6.86) and 0.31 (t 3.91).) The revision of δ between consecutive matched meetings splits into two pieces with different information timing,

$$\Delta\delta_t = \underbrace{dM_t}_{\substack{\text{dealer revision} \\ \text{known before the window}}} - \underbrace{dF_t}_{\substack{\text{dot revision} \\ \text{released inside the window}}},$$

where dM_t is the change in the dealers’ longer-run rate across the two surveys and dF_t the change in the mean dot across the two SEPs. The survey closes before the three-day announcement window $[t-1, t+1]$ opens; the dot is published inside it. Any regression of window changes on $\Delta\delta$ therefore mixes a pre-determined regressor with an announcement-day one, and the split regression — window change on dM and dF entered separately — is the more informative object. This turns out to matter more than power, and it organises everything below: *the announcement window identifies what dF does; the intermeeting period identifies what dM does*. Prices are taken from the ACM decomposition (total, expectations component, term premium, maturities 1–10y) and, model-free, from GSW instantaneous forwards — nominal, TIPS, and inflation compensation, which satisfy nominal = real + breakeven to 10^{-14} on the merged file, so any nominal response splits exactly and without a decomposition step. Two practical points matter. First, the mean of the dots is the usable Fed-side variable: it moves at 47 of 48 meetings, where the median moves at 19 — the earlier finding that the survey-based wedge “gives nothing” was a statement about the median, not about the object. Second, $\text{sd}(\Delta\delta) = 0.117$ pp with $n = 46$ –47, which is the arithmetic behind every power statement below.

5.2 The power calculation Prediction 2 requires

The composition-restriction test regresses far-forward real and breakeven window changes on $\Delta\delta$ and asks whether the coefficient ratio equals $\phi_\pi - 1$. The observed two-day window standard deviations are 10.8 bp for the 5y5y real forward and 6.0 bp for the 5y5y breakeven; with $\text{sd}(\Delta\delta) = 0.117$ and $n = 47$ this gives minimum detectable effects (5% size, 80% power) of 38 bp per pp on the real leg and 21 on the breakeven leg. Against the model’s own effect sizes (the Λ column of §4.5, divided 1:2 between real and breakeven at $\phi_\pi = 1.5$): $\alpha = 0$ implies real ≈ 4 and breakeven ≈ 7 bp per pp — invisible; $\alpha = 0.25$ implies 26 and 51 — the breakeven leg detectable with simulated power 0.99, the real leg not (0.27). The ratio is the casualty: in 5,000 simulated panels the median standard error of $\hat{b}_{\text{real}}/\hat{b}_{\text{bei}}$ is 1.35 at small effects and reaches 0.15 only when the real leg is 40 bp per pp. Separating $\phi_\pi = 1.5$ (ratio 0.5) from $\phi_\pi = 2$ (ratio 1.0) needs a ratio standard error below about 0.25, which requires effects an order of magnitude above the pure-learning prediction.

The conclusion the paper anticipated is now quantitative: Prediction 2 is feasible as a *detection* test — same-direction, breakeven-larger — under moderate α , but it will not identify ϕ_π sharply at the observed variance of $\Delta\delta$. Equation (10)’s “one equation in two unknowns” is, in practice, also one equation in too little variance. The calculation was worth running, but it turned out not to be the binding constraint: §5.5 shows the timing problem of §5.1 does more damage to the literal specification than the standard errors do.

5.3 A fourth fact: the window prices the dot surprise

The restricted regression of window changes on $\Delta\delta$ produces coefficients that are negative, significant, and largest in the belly: the ACM expectations component moves -11.7 bp per pp at 1y ($t-2.4$), -26.0 at 4–5y ($t-3.8$), -22.2 at 10y ($t-3.8$). The sign reads correctly once the timing is kept in mind: $\Delta\delta$ rises when dealers revise up and the dot does not follow, and in exactly those windows yields fall — toward the published endpoint. Unpacking with the split regression shows what the restricted coefficient was averaging (Figure 5, left):

n	b_{dM} (dealer rev.)	b_{dF} (dot rev.)	$b_{dM} + b_{dF}$	$p(b_{dM} = -b_{dF})$
2y	-21.3 ($t-3.5$)	$+22.0$ ($t+2.2$)	$+0.7$ (se 9.5)	0.94
5y	-26.0 ($t-3.6$)	$+25.8$ ($t+2.7$)	-0.2 (se 10.0)	0.99
10y	-22.2 ($t-3.6$)	$+23.0$ ($t+2.9$)	$+0.9$ (se 8.3)	0.92

The two coefficients are equal and opposite at every maturity — the equality is never rejected, $p = 0.92\text{--}0.99$ — so the window prices the single variable $dF - dM$: *the part of the dot revision that the dealers’ own preceding revision did not already contain*. This is the pricing counterpart of Fact 2. There, dealer revisions load on the preceding and contemporaneous dot revisions (0.60, 0.58), so the survey is, among other things, a forecast of the dot; here, the market’s response to the announcement is measured against precisely that forecast. It is also a discipline on Fact 1’s interpretation: the window moves attributed to Δi^* are not responses to the published revision per se, but to its innovation relative to a measurable prior expectation — an expectations-versus-realization structure that the Δi^* reading implicitly assumes and the split regression now verifies. Robustness: the pattern is unchanged in the one-day window ($-18.9/+29.8$), in both the 2013–2019 and 2020–2026 halves, and with median dots the dM side survives while the dF side weakens ($+12.7, t1.5$) — the median’s 19 moves are simply too few.

5.4 Prediction 4: the endpoint pass-through is half, and not flat

Prediction 4 says a revision to the published endpoint loads flat, at one, across maturity. The model-free version of the test regresses GSW instantaneous-forward window changes on dM and dF ; the dot-revision loading $b_{dF}(n)$ is the estimated endpoint pass-through (Figure 5, centre). It is $+47$ bp per pp at the 1y forward ($t2.8$), rises to $+74$ at 4–5y ($t5.2\text{--}5.4$), and falls to $+49$ at the 10y forward ($t2.4$). Two statements follow, with different force. The *level* is rejected: at the ten-year forward the pass-through is 0.49 (se 0.21), $t = -2.5$ against one. Half of a mean-dot revision reaches the far forward curve; the model’s full pass-through overstates the long-end response by a factor of two. The *shape* is suggestive but not established: the individual contrasts $b_{dF}(4y) - b_{dF}(1y) = +27$ ($t1.6$) and $b_{dF}(10y) - b_{dF}(4y) = -26$ ($t-1.3$) do not reach significance, so a hump peaking at 4–5y is the point estimate, not a finding. Two readings are consistent with the sub-unit level: the market treats part of a dot revision as transitory — the Fed-learning discount of $\$4.5$ applied to the Fed’s own announcements — or the mean dot moves at meetings where part of the revision was already priced through other communication. One further detail from the forwards is worth recording: beyond the 4y forward the surprise restriction $b_{dM} = -b_{dF}$ is rejected ($p < 0.01$) —

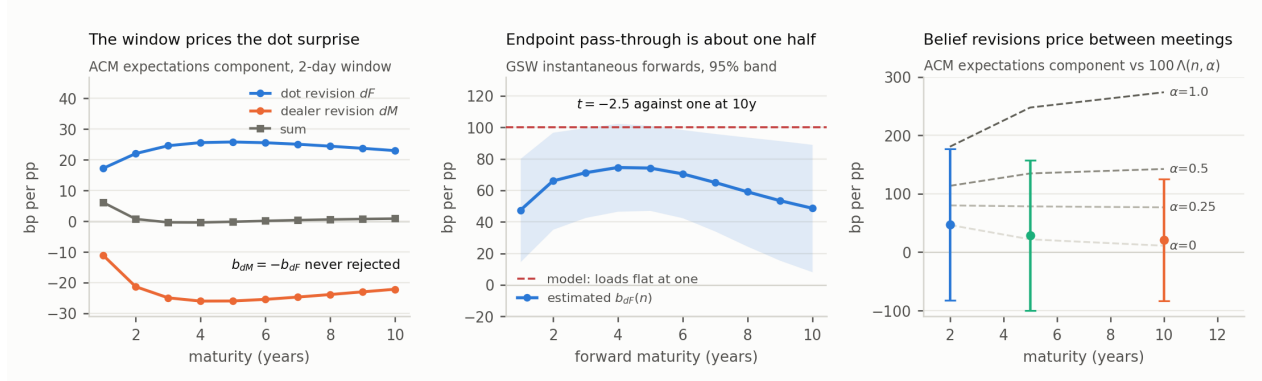


Figure 5: The dot revision and its pricing. Left: split regression of two-day window changes in the ACM expectations component on the dealer revision dM (pre-window) and the dot revision dF (in-window), by maturity; the coefficients are equal and opposite ($p = 0.92\text{--}0.99$), so the window prices the dot surprise $dF - dM$, and their sum is zero. Centre: the dot-revision loading across GSW instantaneous forwards with 95% bands; Prediction 4’s flat-at-one loading is drawn for reference and rejected at the far end. Right: intermeeting pricing of dealer revisions in ACM expectations components against the model loadings $100\Lambda(n, \alpha)$. HAC(4); 46–47 matched meetings, 2013–2026.

distant forwards respond to the dot revision beyond what the dealer revision anticipated — while in ACM expectations space it never is. The ACM expectations component is a filtered object with compressed far-forward variation (the stationary-VAR critique documented in the archive), so where the two disagree, the raw forwards carry the information.

5.5 Prediction 2: the composition restriction

The TIPS curve makes the composition test operational. Its attraction is that the ratio is *free of* α : the real and breakeven loadings are $\Lambda(n)(\phi_\pi - 1)/\phi_\pi$ and $\Lambda(n)/\phi_\pi$, so the common Λ cancels and the ratio is $\phi_\pi - 1$ whatever the permanent share. This makes Prediction 2, not Prediction 5, the cleanest α -free content of the framework. But the ratio is a ratio *of responses to a movement in* δ , and by §5.1 the announcement window does not deliver one: a dot revision moves the published endpoint and δ at the same instant. Three specifications separate the cases.

(A) The composition of a dot revision. Regress window changes on dF . Since π^* is 2.0 in every SEP, a dot revision *is* a real endpoint revision, and the model predicts it arrives almost entirely in the real leg, with a small negative breakeven response from the induced fall in δ : $b_{\text{real}} = 1 - (1 - \theta)\lambda(\phi_\pi - 1)/\phi_\pi$ and $b_{\text{bei}} = -(1 - \theta)\lambda/\phi_\pi$, where θ is the share of the revision the market adopts into its own belief inside the window. The data (Figure 6, left) are emphatic:

forward	real (TIPS)	breakeven	nominal
2y	+42.4 (t 2.32)	−11.9 (t −1.40)	+30.5 (t 2.23)
5y	+53.8 (t 3.90)	+1.2 (t 0.17)	+55.0 (t 4.47)
10y	+44.7 (t 4.82)	−0.1 (t −0.01)	+44.6 (t 3.91)
5y5y	+51.7 (t 5.03)	+0.9 (t 0.10)	+52.3 (t 4.97)

The half pass-through of §5.4 is entirely real. Breakevens do not move on a dot revision at any forward maturity beyond three years; the largest breakeven coefficient anywhere on the curve is −11.9 bp per pp at the 2y forward, t −1.40. This is Hillenbrand’s composition result — the decline is in TIPS, not breakevens — reproduced at the meeting frequency and on the published projection

specifically, and it is the sharpest available evidence that the market reads the SEP longer-run dot as a statement about the real endpoint rather than about the inflation anchor. That is the interpretation (13) requires, and it had not been tested directly.

(B) The composition of a disagreement. The ratio test needs a movement in δ that does not touch the published endpoint, which is what dM is. Regressing the intermeeting change — the close two days before meeting t against the close after meeting $t - 1$ — on dM (Figure 6, centre):

forward	real	breakeven	ratio	Fieller 95% (ratio)	implied ϕ_π
9y	+99.1 (t 2.24)	+52.9 (t 2.44)	1.87	[0.66, 3.52]	[1.66, 4.52]
10y	+95.6 (t 2.16)	+55.1 (t 3.02)	1.74	[0.33, 2.86]	[1.33, 3.86]

Three things to take from this, in descending order of confidence. First, **the two legs move together**, which is the model’s qualitative requirement and the feature that demand disagreement does not generate; the contaminated specifications in (A) and (C) both show opposite signs, so this is not an artefact of the data always co-moving. Second, the point estimate, $\phi_\pi \approx 2.7$, is higher than the 1.5 used for calibration, and its interval comfortably contains 1.5; it should be read as a first estimate rather than a number to build on. Third, and most tentatively, the ratio is probably positive, which would put ϕ_π above one — the check §4 demands of the empirical work. The evidence for that is weaker than the Fieller intervals suggest: resampling the panel directly gives $\Pr(\text{ratio} > 0) = 0.97$ (iid) and 0.95 (moving block, $L = 6$), with percentile intervals of $[-0.17, 4.11]$ and $[-0.78, 4.00]$ that include zero. The Taylor principle is favoured by the data at roughly one-sided 95%, not confirmed at conventional two-sided levels. Intervals are Fieller rather than delta-method because the denominator is imprecise — at every forward maturity except 9 and 10 years the 95% set is unbounded, which is exactly the failure mode §5.2 predicted, and a reason not to lean on the two maturities where it happens to close. The caveat is the design, not the arithmetic. An intermeeting window is six to eight weeks of macro news, and dealers revise r^{*m} in response to the same news that moves real rates and inflation compensation directly; this is a correlation with the right sign and the right ordering, not an identified response. Conditioning on the intermeeting move in the 2y and 3y nominal forwards — the nearest available proxy for the common cycle — leaves the ratio at 1.81 but widens the interval to include zero, and dropping 2020 leaves 1.83 with an interval of $[-0.08, 3.22]$. The composition restriction is therefore *consistent with* the data and now has a first estimate; it is not yet established.

(C) The literal specification. Regressing window changes on $\Delta\delta$, as §4.7 originally proposed, gives -14.1 (t -0.96) real and $+4.8$ (t 0.51) breakeven at the 10y forward, and -65.4 (t -3.24) and $+19.0$ (t 1.61) at 2y: opposite signs, which the model forbids, and a ratio near -3 . This is not evidence against the model. It is the timing problem: the window regression on $\Delta\delta$ is dominated by dF , whose composition is the near-pure real response of (A), so the specification measures the composition of an endpoint revision and reports it as though it were the composition of a disagreement. The lesson for the draft is that Prediction 2 must be stated on dM and estimated between meetings, not on $\Delta\delta$ in the window.

5.6 Bounding the permanent share α

The amplification term $\Lambda(n)\delta_t$ is the model’s channel from disagreement to the level of yields, and §4.5 showed its magnitude turns entirely on α . The far-forward data bound it from both sides. The instrument is (11) evaluated at a long horizon rather than averaged. An instantaneous forward

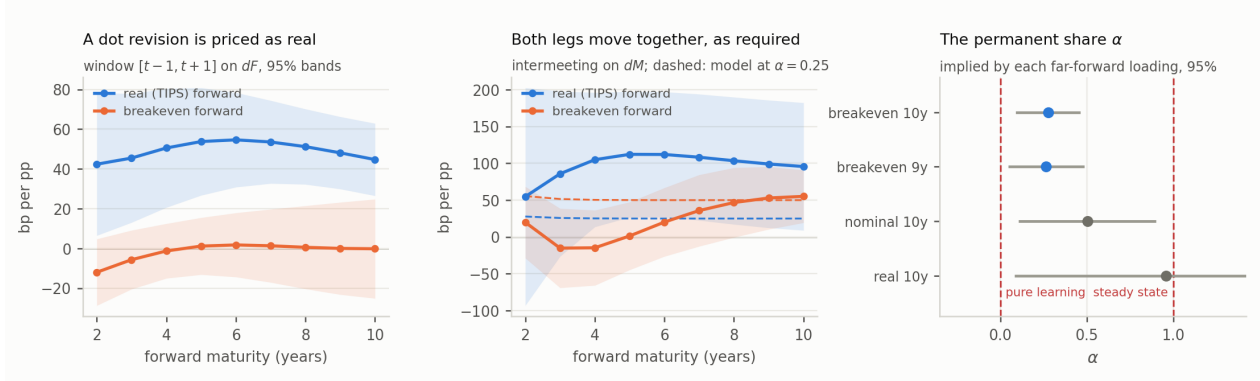


Figure 6: The composition restriction. Left: window changes in GSW real (TIPS) and breakeven instantaneous forwards regressed on the dot revision dF — the response is entirely real at every maturity beyond three years. Centre: the same two legs regressed on the dealer revision dM between meetings, with the model’s predicted loadings at $\alpha = 0.25$ dashed; both legs move together, as the model requires and as the window specifications do not. Right: the permanent share α implied by each far-forward loading, where $\lambda_{120} \rightarrow 3\alpha$ makes the ten-year instantaneous forward a near-pure reading of α ; $\alpha = 0$ and $\alpha = 1$ are marked. HAC(4), 95% intervals; Fieller intervals for ratios.

j months ahead loads λ_j , and at $j = 120$ the transient factor $(1 - \mu)^j = 0.897^{120} = 2 \times 10^{-6}$ is dead, so

$$\lambda_{120} \rightarrow \frac{\phi_\pi}{\phi_\pi - 1} \alpha = 3\alpha,$$

and the real and breakeven legs load 100α and 200α bp per pp. *The ten-year instantaneous forward is a near-pure reading of the permanent share*, and the breakeven leg is the best-measured of the three, and the only one whose interval is stable under resampling (moving-block bootstrap median 0.25, 95% [0.05, 0.45], $\Pr(\alpha > 0) = 0.99$). From the intermeeting regressions of (B):

	coefficient	implied α	95% interval
10y forward, breakeven	+55.1 (se 18.2)	0.28	[0.10, 0.45]
9y forward, breakeven	+52.9 (se 21.7)	0.26	[0.05, 0.48]
10y forward, nominal	+150.8 (se 59.8)	0.50	[0.11, 0.89]
10y forward, real	+95.6 (se 44.2)	0.96	[0.09, 1.82]

The breakeven reading survives the same robustness that weakened the ratio: $\alpha = 0.25$, [0.06, 0.43] excluding 2020, and $\alpha = 0.19$, [0.02, 0.37] conditioning on the 2y and 3y nominal forward moves. So a one-sentence summary of the amplification mechanism is now available: *α is about a quarter, and both polar cases are rejected.* The steady-state case $\alpha = 1$ — the multiplier of (10) taken at face value — is excluded by every column. Pure learning, $\alpha = 0$, is excluded by the breakeven columns, which is the more interesting of the two rejections: the calibration of §4.5 treated $\alpha \approx 0$ as the case that would kill the magnitude claim, and it does not hold. At $\alpha = 0.25$ the ten-year endpoint multiplier is $\Lambda = 0.77$: a 10 bp disagreement moves the ten-year expectations component about 8 bp, against 1 bp under pure learning and 27 bp in the steady state. Two consistency checks. The announcement-window evidence gives the same upper bound from an independent direction. In the window, dealer belief revisions are pre-determined, and the sum $b_{dM} + b_{dF}$ of §5.3 would capture Λ only if beliefs reached prices solely at announcements; it is zero everywhere (+0.9 bp per pp at 10y, se 8.3, block-bootstrap 95% CI [−14.9, +17.8]), which is what continuous pricing predicts and therefore tells us nothing about α by itself. The window breakeven response to dF is more informative, but only conditionally: it bounds $(1 - \theta)\alpha$, where θ is in-window belief adoption.

At $\theta = 0$ it gives $\alpha \leq 0.13$; at $\theta = 0.5$, $\alpha \leq 0.25$; at $\theta = 0.82$ — the Fact-2 survey-to-survey adoption coefficient, the most generous case — $\alpha \leq 0.70$. Since θ is unobserved, this is a range that brackets the intermeeting estimate rather than an independent test, but it does not contradict it. And the ACM expectations components at the yield level, regressed on dM intermeeting, give $\alpha \leq 0.43$ at ten years (Figure 5, right) — an upper bound from a filtered, model-based measure that lands within 0.02 of the raw-forward interval’s upper end. Three measurements, two of them independent, agree on the ceiling. For the draft this resolves the instruction of §4.5 in the least drastic direction available: the magnitude claim need not be dropped, but it must be restated at $\alpha \approx 0.25$ rather than at the steady state, where it is roughly a quarter of the headline multiplier and still economically meaningful. The peak-location test (12) remains unrun — it presupposes a Λ profile precise enough to locate a maximum, and the intermeeting standard errors do not support one. The hump that *is* in the data (§5.4) belongs to the dot-surprise loading, a different object, with a peak at 4–5y rather than the 1–2.5y the Λ family predicts.

5.7 Prediction 3: symmetry survives

Splitting the level response at $\delta = 0$ (10y, two-day window): $p(\text{symmetric}) = 0.98$ for the total yield, 0.79 for the expectations component, 0.87 for the term premium. The model’s linearity — the feature that distinguishes it from Caballero & Simsek’s sign-dependent premium — is not rejected, and the earlier appearance of asymmetry in the ACM-based wedge does not reappear in the survey-based measure, consistent with its being the regime artefact the gate analysis suspected. One flag for completeness: in revisions, the ten-year term-premium response is asymmetric (extra slope for $\Delta\delta > 0$ of -65 , $t - 2.1$, $p = 0.035$) — one rejection among ten symmetry tests, on 46 observations, worth re-checking when the sample grows rather than interpreting now.

5.8 Prediction 6: nothing on the available proxy

The disagreement premium requires $\sigma_{\delta,t}^2$, which no survey measures. The nearest available object is the dealers’ cross-sectional IQR of the longer-run rate (91 surveys, mean 0.45, AR(1) 0.58). ACM term premia show no sensitivity to it at any maturity — levels flat with $|t| < 0.9$ everywhere, changes negative and insignificant, and no rise with maturity, which is the signature the prediction needs. The caveat cuts both ways: forecaster dispersion is dispersion *within* the market, not uncertainty about the Fed–market gap, so this is a test of a proxy, not of the prediction. But it removes the easy version of the result: whatever Cao, Crump, Eusepi & Moench’s dispersion factor is picking up in term premia, the dealers’ longer-run IQR alone does not reproduce it on this sample.

5.9 Where this leaves the program

The empirical section of the paper changes in five ways. Fact 2 gains its pricing counterpart — the fourth fact of §5.3 — which is also the cleanest evidence yet for the anchoring claim of (13): yields are organised around the Fed’s published endpoint, to the point that the announcement response is a pure surprise response with the dealers’ survey as the measurable expectation. Prediction 4 acquires a number: the far-forward pass-through of a dot revision is one half, not one, and §5.5 shows that half is entirely real — which both confirms the framework’s reading of the SEP dot as a real-endpoint statement and reproduces Hillenbrand’s TIPS-versus-breakevens composition on the published projection itself. Prediction 2 becomes operational, with the caveat that it must be estimated between meetings rather than in the window; it delivers the model’s qualitative requirement (both legs, same sign), a first ratio estimate implying $\phi_\pi \approx 2.7$, weak one-sided support for $\phi_\pi > 1$, and a clear statement of what would sharpen it. The amplification mechanism is disciplined to $\alpha \approx 0.25$ with a 95% interval of roughly $[0.10, 0.45]$, so both the steady-state multiplier and pure transience are rejected and the ten-year endpoint loading is about 0.8. And Prediction 6 remains genuinely open, awaiting a credible measure of σ_δ^2 rather than a dispersion proxy. Two vulnera-

bilities are worth stating in the same breath as the results. The intermeeting design that carries Prediction 2 and the α bound cannot separate the model’s mechanism from common macro news; an instrument for dealer belief revisions, or a higher-frequency survey, is the natural next step. And the sample is what it is: $\text{sd}(\Delta\delta) = 0.117$ on 46 meetings, so everything here is a statement about small revisions to a small gap, and the program’s power grows only at eight observations a year.

Key references. Bauer, Pflueger & Sunderam (2024), *QJE*. Bauer & Rudebusch (2020), *AER*. Caballero & Simsek (2022), *AER*. Cao, Crump, Eusepi & Moench, FRBNY SR 934. Cho & Williams (2026), Liberty Street Economics. Cieslak, McMahon & Pang, SSRN 4920858. Engstrom, FEDS 2026-026. Gürkaynak, Sack & Swanson (2005). Gürkaynak, Sack & Wright (2007; 2010), *JME*; *AEJ:Macro*. Hanson, Lucca & Wright (2021), *QJE*. Hartley (2025), Mercatus WP. Hillenbrand (2025), *RFS*. Holston, Laubach & Williams (2017), *JIE*; (2023), FRBNY SR 1063. Joslin, Singleton & Zhu (2011), *RFS*. Kozicki & Tinsley (2001), *JME*. Lou, Pintér, Üslü & Walker, BIS WP 1232. Lucca & Moench (2015), *JF*. Orphanides (2003), *JME*. Orphanides & Williams (2002), *BPEA*. Pan & Peng, SSRN 4764451. Swanson (2021), *JME*.